MIL-T-45917A
4 March 1974
SUPERSEDING
MIL-T-45917
18 March 1968
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20 May 1965

MILITARY SPECIFICATION

TARGET MECHANISM, NIGHT FIRING, SMALL ARMS, M40

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of electro-mechanical target mechanism training device, designed to operate on a 115 volt, 60 cycle, A. C. power source (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-W-12332	-	Welding, Resistance, Spot, Seam and Projection; for Fabricating Assemblies of Low-Carbon Steel
MIL-P-14232	-	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of

STANDARDS

Federal

FED-STD-595	-	Colors

FSC 6920

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Military

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-171	-	Finishing of Metal and Wood Surfaces
MIL-STD-454	-	Standard General Requirements for Electronic Equipment
MIL-STD-1261	-	Welding Procedures for Constructional Steels

DRAWINGS

Rock Island Arsenal

F7545932	-	Target Mechanism, Night Firing, Small Arms, M40
F7545933	-	Hit Indicator Assembly
C7545995	-	Hit Switch Assembly
F12002898	-	Target, Silhouette, Type "E" Kneeling, Polyethylene

PUBLICATIONS

Rock Island Arsenal

P7545932	-	Packaging Data Sheet for Target Mechanism, Night Firing, Small Arms, M40

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ANSI Y32.3 (AWS A2.0-58) - Welding Symbols

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10018.)

3. REQUIREMENTS

3.1 Material. Material shall conform to the material specifications referenced on the detail drawings of Drawing F7545932.

3.2 Dimensions and tolerances. Dimensions and tolerances of parts and assemblies shall conform to the detail drawings of Drawing F7545932.

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3.3 Construction.

3.3.1 Assembly. Parts and assemblies of the target mechanism shall be positioned and attached in accordance with the applicable drawings of Drawing F7545932.

3.3.2 Welding. Spot-welding and welding of joints of the target mechanism shall be in accordance with the specifications referenced on the applicable drawing. Interpretation of welding symbols shall be as defined in ANSI Y32.3, Welding Symbols. Actual production of weldments shall not proceed prior to Government approval of the contractor's welding procedures and test specimens.

3.3.3 Soldering. The soldering of electrical connections shall be in accordance with Requirement 5 of MIL-STD-454.

3.4 Protective finishes.

3.4.1 Inorganic finishes. Inorganic finishes shall be in accordance with the Finish Number of MIL-STD-171 specified on the applicable detail drawings of Drawing F7545932.

3.4.2 Painting. The Target Mechanism and parts shall be painted in accordance with the Finish Number of MIL-STD-171 specified on the drawings. Parts not to be painted shall be in accordance with the same standard. Unless otherwise specified in the contract or order, the Finish Number of MIL-STD-171 to be used for cleaning shall be optional with the contractor. Color shall be as specified by the procuring activity (see 6.2).

3.5 Performance and product characteristics.

3.5.1 Electrical system. The electrical system for the Target Mechanism shall have a power source of 115 volts, plus or minus 5 volts, 60 cycle, alternating current.

3.5.2 Flasher assembly. When the selector switch is moved to the "Up" position for single fire, the target flasher light selected shall flash once every 3 seconds, plus 2 seconds.

When the selector switch is moved to the "Down" position for rapid fire, the target flasher light selected shall flash in a burst of 3 to 5 times, at intervals of 3 seconds, plus 2 seconds.

The switch strip (switch identification) shall be illuminated through the plastic strip by 2 panel lights, one on each end. Each flasher switch shall cycle in all 3 positions. The device shall be capable of carrying the electrical load of all 15 circuits at one time.

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3.5.3 Hit indicator assembly. When the hit switch is actuated, the corresponding counter shall record the actuation and the corresponding panel light shall go out and relight after the impulse.

3.5.4 Performance. With flasher lights emplaced and operating as specified in 3.5.2, the mechanism shall be capable of performing the following functions at a distance of 50 yards in an outside area using caliber .30 or 7.62MM ammunition.

Record one burst of 5 rounds of automatic fire and 5 single rounds of semi-automatic fire at each of the 15 silhouettes. There shall be no more than 3 "No Kills" for each 75 rounds and no more than one "No Kill" shall be allowed per each hit indicator circuit.

3.6 Marking. Marking of the target mechanism and component parts shall be in accordance with the applicable drawings and MIL-STD-130 (see 6.3.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions as used in this specification are in accordance with MIL-STD-109.

4.3 Preliminary inspections. Prior to initiating acceptance inspection, one completed target mechanism, taken at random from the initial lot, shall be subjected to all examinations and tests specified herein. If that mechanism fails to pass all the examinations and tests, acceptance inspection shall be deferred pending corrective action by the contractor. After Government verification of corrective action by the contractor, the examination and tests shall be repeated on another mechanism (other than the one originally tested). Acceptance inspection shall be initiated after the mechanism has passed this inspection.

4.4 Inspection provisions.

4.4.1 Materials. Materials used in construction of the target mechanism shall be tested in accordance with the material specifications referenced on the applicable drawings (see 3.1).

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4.4.2 Welding. Inspection, testing and acceptance of weldments and welding procedures shall be in accordance with MIL-W-12332 and MIL-STD-1261 referenced on the applicable drawings (see 3.3.2).

4.4.3 Soldering. Inspection for soldering requirements shall be in accordance with Requirement 5 of MIL-STD-454.

4.4.4 Protective coatings. Examination and test of inorganic finishes and painted surfaces shall be in accordance with the specifications specified on the applicable drawings and MIL-STD-171 (see 3.4.1 and 3.4.2).

4.5 Sampling for lot acceptance.

4.5.1 Lot. A lot shall consist of all target mechanisms manufactured under essentially the same conditions and submitted at the same time.

4.5.2 Sampling for examination. A random sample of completed target mechanisms shall be taken from each lot in accordance with MIL-STD-105, at Inspection Level II. Acceptable Quality Levels (AQL) shall be in accordance with Table I.

4.5.3 Sampling for tests.

4.5.3.1 Flasher assembly, electrical load, hit indicator assembly. One sample, picked at random from each 25 target mechanisms manufactured, shall be tested in accordance with 4.7.2, 4.7.2.1 and 4.7.3.

4.5.3.2 M40 mechanism operation. One sample mechanism taken at random from each 100 completed target mechanisms shall be tested in accordance with 4.7.4.

4.5.4 Sampling of packaging and packing. Sampling for inspection of packaging and packing of the target mechanism shall be as specified in MIL-P-14232.

4.6 Examination.

4.6.1 Each sample mechanism taken in accordance with 4.5.2 shall be examined to determine conformance to this specification. Examination shall be conducted in accordance with Table I.

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Table I

Categories	Defects	Inspection Method
Critical	None defined	
Major	AQL = 0.65 percent defective	
101	Incorrect assembly or missing parts	Visual
102	Welding not as specified	Visual
103	Soldering not as specified	Visual
104	Circuit defect: Discontinuity (open circuit) between pin of connectors and contacts	Measure
105	Circuit defect: Continuity (short circuit) from any pin connectors or box	Measure
106	Discontinuity or absence of protective coating	Visual
Minor	AQL = 4.00 percent defective	
201	Dimensional characteristics	Measure
202	Color of paint	Visual
203	Discontinuity or absence of paint	Visual
204	Loose or unconnected wires	Visual
205	Failure of lamps to light	Visual
206	Incorrect markings	Visual
207	Incorrect or missing part number	Visual

4.6.2 Packaging and packing. Inspection of packaging and packing of the target mechanism shall be in accordance with MIL-P-14232.

4.7 Test procedures.

4.7.1 Equipment. Each target mechanism taken for test (see 4.5.3) shall be equipped with the following parts:

a. Fifteen silhouette targets, F12002898
b. The targets shall be installed on the target holder assemblies.
c. Cables, cords, etc., needed for normal operation and the mechanism connected to a power source as specified in 3.5.1.

4.7.2 Flasher assembly. Using an appropriate timing device, the number of flashes shall be measured for both the "Up" position (single fire) and "Down" position (rapid fire) on each of the 15 selector switches. Failure to comply with the requirements of 3.5.2 shall be cause for rejection of the lot.

4.7.2.1 Electrical load. Each of the 15 flasher control switches shall be activated to the "Up" position, then the "Down" position. The mechanism shall be capable of carrying the electrical load of all 15 circuits

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in either position at one time. Failure to pass this test shall be cause for rejection of the lot.

4.7.3 Hit indicator assembly. With the Hit Switch Assemblies C7545995 connected to the Hit Indicator Assembly F7545933, each of the 15 hit switch assemblies shall be actuated to determine conformance to 3.5.3. Failure to pass this test shall be cause for rejection of the lot.

4.7.4 Performance. Equipped as specified in 4.7.1, and with flasher lights emplaced and operating, the mechanism taken in accordance with 4.5.3.2 shall be subjected to 5 rounds of automatic fire and 5 rounds of semiautomatic fire to determine conformance to 3.5.4. Failure of the mechanism to pass this test shall be cause for rejection of the lot.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. Preservation, packaging, packing and marking of the target mechanism shall be in accordance with Packaging Data Sheet P7545932.

5.2 Level. The level of protection required shall be as specified in the contract or order (see 6.2).

6. NOTES

6.1 Intended use. Target Mechanism M40 is used for training in night firing of small arms. The target mechanism permits the operator to control fifteen target locations and will accommodate either Kneeling Target F12002898 or Prone Target D12002899.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Title, number and date of applicable drawings.
c. Level of protection required.
d. Color (see 3.4.2 and 6.3.2).

6.3 Suggested contractual features.

6.3.1 Serial numbers. The contractor should be furnished a block of serial numbers by the Contracting Officer to be used for marking the mechanism nameplate.

6.3.2 Colors. Color Chip Numbers 24087 and 37038 of Fed. Std. 595 may be used as a guide for painting.

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6.3.3 Government furnished equipment. An adequate number of Targets, Kneeling, F12002898, should be furnished the contractor for test purposes (see 4.7.1).

Custodians:
Army - WC
Navy - TD
Air Force - 70

Review activities:
Army - None
Navy - None
Air Force - None

Preparing Activity:
Army - WC

Project No. 6920-0070

User activities:
Army - None
Navy - None
Air Force - None

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-T-45917A, Target Mechanism, Night Firing, Small Arms, M40

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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